RUSHI ASHOK SAVANI

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EDUCATION

Syracuse University, NY, USA **Aug 2024 - May 2026**
Masters | Information Systems, Data Science (GPA: 3.86/4)

- *Relevant Coursework:* Applied Machine Learning, Database Management, Data Science, Cloud Management

Thadomal Shahani Engineering College, Mumbai, India **Aug 2020 - Jun 2024**
Bachelors | Information Technology *(Minors - Artificial Intelligence and Data Science)* (GPA: 8.52/10)

WORK EXPERIENCE

AI Engineer / Data Science Co-op **Jun 2025 – May 2026**
ProHealth Home Health and Hospice *Atlanta GA*

- Built a 4-agent **LangGraph** pipeline in **Snowflake** with Cortex Search, session memory, and retry checks for fragmented patient data, and cut administrator workload by 20+ hours weekly.
- Audited logs against a 50+ question eval set built with the manager, traced failures to semantic retrieval, and fixed it with hybrid retrieval, lifting accuracy 35% and answers to 90%.
- Rebuilt an **XGBoost** readmission model with **SHAP**-guided feature selection, resolving class imbalance and multicollinearity, and lifted recall from 48% to 76% as readmissions fell **10%** with care team follow-up.
- Designed **3 LLM** pipelines generating citation-backed root cause reports for flagged cases clinicians could not manually investigate, which surfaced about **$20K** in preventable opportunities during Q1.
- Selected models through A/B testing and LLM-as-judge scoring against a custom rubric with manual review, then kept prompts stable via DeepEval's hallucination and factual-consistency checks.
- Deployed SageMaker endpoints and the investigation pipeline via Dockerized FastAPI on **AWS EC2** with GitHub Actions CI/CD, and CloudWatch monitoring caught a 10% recall drift in production with zero downtime.

Data Scientist - Research Assistant **Jan 2025 – May 2025**
Syracuse University *Syracuse, NY*

- Automated a **Python** pipeline that handled missing values, standardized unstructured text and normalized mixed language responses into English before fine-tuning, saving 15+ hours.
- Fine-tuned RoBERTa in **PyTorch** via Hugging Face Transformers on ~7,000 survey responses, tuning optimal learning-rate through loss curve with AdamW, reaching **91%** accuracy under stratified K-fold validation.
- Clustered open responses using Sentence-BERT embeddings and HDBSCAN, then exposed AI adoption themes through a Streamlit dashboard built for non-technical research stakeholders.

Data Analyst **Jun 2023 - Jun 2024**
Krishna Jewels *Mumbai, IN*

- Migrated 40K+ paper based inventory, sales, and supplier records into a normalized **MySQL** schema, cutting record lookup time by 70% through indexing and query rewrites.
- Automated 15+ hours of weekly manual analysis by building SQL ETL pipelines into **Power BI** with DAX measures for sales, margin, and inventory KPIs, building interactive dashboards.

TECHNICAL SKILLS

- **Languages & Data:** Python, SQL, PostgreSQL, MySQL, TypeScript, Git, REST APIs
- **LLM & GenAI:** LangGraph, RAG, Vector Databases, Prompt Engineering, DeepEval, RAGAS
- **Machine Learning:** Scikit-learn, NumPy, XGBoost, Classification, Regression, MLflow, SHAP, Cross Validation, A/B Testing
- **Deep Learning & NLP:** PyTorch, TensorFlow, Hugging Face Transformers, RoBERTa, ClinicalBERT, Sentence-BERT, TF-IDF
- **Analytics & BI:** Power BI, Tableau, Streamlit, ETL/ELT, Data Modeling, Statistical Analysis, Hypothesis Testing
- **MLOps & Cloud:** AWS SageMaker, Snowflake, Docker, FastAPI, GitHub Actions, CI/CD, CloudWatch, Model Monitor

PROJECTS

PrivateRAG | *Fully Local Private-Document Q&A System, GitHub*

- Designed a fully local **Retrieval-Augmented Generation (RAG)** stack with document ingestion, Qdrant vector indexing, metadata filtering, and Llama 3.1 inference with zero external API calls.
- Wrote a TypeScript chat interface over a local OpenAI-compatible backend with streaming responses, source attribution, and chunk-level traceability.

COVID-19 India Analytics | *Interactive Tableau Dashboard, Dashboard*

- Visualized India's COVID-19 case, testing, and vaccination trends in one interactive **Tableau** dashboard.
- Wrangled several large public datasets in Python, reconciling inconsistent state-level records before loading them into Tableau for analysis.
- Mapped vaccine dose administration against testing capacity, exposing regional disparities across age groups

KarmaScore Predictor | *Subscription Churn Analysis*

- Trained an XGBoost churn classifier on 30K+ subscriber records, narrowing 60 candidate variables to 23 predictors through recursive feature elimination and mutual information scoring.
- Ranked subscribers into decile risk bands using PR-AUC and lift analysis, producing a prioritized list for targeted retention planning.